

Ethan Brann

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Experience

Bluebrick.ai

Software developer

Aug 2024 - present

- Helped develop generative algorithms used to create feasibility studies for construction projects
- Led integration of 3D map tiles from Google Maps API into application, including writing custom shader for seamless integration.
- Prototyped and implemented ChatGPT function calling feature to create text-based interface for application.

Freelance software development

Feb 2024 - Aug 2024

- Worked closely with CEO of an AR/3D startup to develop MVP for a new product. Designed and built a system that generates and renders a 3D scene to video based on a JSON config file.
- Built digital catalogue for new property development with an emphasis on slick user experience. Completed work ahead of time to a high standard.

Modulous

ThreeJS developer

March 2023 - Dec 2023

- Collaborated in the design and development of a suite of algorithms that generated building designs in adherence to current building rules and regulations.
- Worked closely with subject matter experts, such as architects, to understand the product requirements and ensure the product fulfilled its intended purpose.
- Contributed to the development of a rendering pipeline that rendered the output of generative algorithms (React, three.js, react-three-fiber). Specifically adding the ability to apply textures to generated meshes.

Poplar Studio

Full stack developer

Aug 2020 - Feb 2023

Junior developer

Nov 2019 - Aug 2020

- Collaborated effectively within a small cross-functional team consisting of a designer, product owner, ML engineer and fellow developers.
- Contributed to on every layer of the tech stack, taking on responsibilities such as implementing frontends (React, Vue.js), designing and implementing REST APIs (node.js, express.js), and provisioning infrastructure (AWS, GCP, Docker).
- Led the R&D and subsequent implementation of a browser-based 3D model viewer (three.js), seamlessly integrating it into various products.
- Organised and participated in hack days, creating prototypes in areas such as immersive audio, 3D file conversion, PBR rendering and more.
- Worked closely with ML engineer to ready a computer vision pipeline for use in a product.

Education

Bath Spa University

(2014 - 2018)

2:1 BA (Hons) Creative Music technology

- Studied modules such as Digital Signal Processing, Audio software development
- Implemented phase vocoder using Pure Data
- Programmed a VST/AU stereo delay plugin using C++ and the Juce framework

- Used Wwise to add sound effects to a scene in Unity
- Created a particle based audio visualiser using Max/MSP

Projects

WebAudio drum machine

<https://www.ethanbrann.co.uk/drum-machine/>

- Browser based application aiming to capture the feel of classic drum machines such as Roland 808 and 909
- Uses WebAudio API to create sequencer and playback audio samples at varying speeds and volumes
- Designed and engineered 100% by myself

Skills

Languages - TypeScript, JavaScript, HTML, CSS, Sass, PHP

Frontend - React, Vue, Three.js, Web Audio API

Backend - Node.js, Express.js, Laravel

Tools/DevOps - Git, Webpack, Vite, Docker, Kubernetes, AWS, GCP, Github

Testing - Jest, Vitest, Cypress

Database - Postgres, Sqlite, DynamoDB